

# Course Outline: Python Data Types and Operators

**Title:** Python Data Types and Basic Operators **Learnpack:** + Tipos de datos y operadores básicos de Python **Prerequisite:** Intro to Python (variables, syntax basics) **Target Audience:** Beginner developers transitioning from JavaScript/TypeScript to Python **Total Lessons:** 18 **Estimated Total Length:** ~9,000 words **Format:** Mixed (33% hands-on)

## Course Description

This course introduces Python's primitive data types and fundamental operators. Students will learn to identify, distinguish, and work with integers, floats, booleans, strings, and the None type. The course builds on the variable concepts from the previous learnpack and prepares students for control flow structures.

By drawing parallels to JavaScript (which students already know), this course emphasizes recognizing data types by their composition and understanding when to use each type. Students will practice arithmetic, assignment, comparison, and logical operators through hands-on challenges that reinforce pattern recognition over memorization.

The course follows 4Geeks' path of least resistance philosophy: learn the essentials first, practice immediately, and build confidence through repetition. No AI assistance is permitted—students must write and understand their own code.

## Course Philosophy

- This course emphasizes:
- **Association over memorization:** Connect Python concepts to JavaScript knowledge you already have
  - **Type recognition by value:** Learn to identify data types by looking at their composition
  - **Practical application:** Every concept is followed by hands-on practice
  - **Research skills:** Develop the habit of looking things up instead of memorizing syntax

## Course Statistics Summary

| Section                                       | Lessons | Estimated Words |
|-----------------------------------------------|---------|-----------------|
| 00 - Welcome                                  | 1       | ~450            |
| 01 - Understanding Primitive Data Types       | 4       | ~2,100          |
| 02 - The None Type and Type Detection         | 3       | ~1,600          |
| 03 - Arithmetic and Assignment Operators      | 3       | ~1,600          |
| 04 - Comparison and Logical Operators         | 3       | ~1,600          |
| 05 - Common Mistakes with Types and Operators | 2       | ~1,050          |
| 06 - Assessment                               | 1       | ~700            |

| Section         | Lessons | Estimated Words |
|-----------------|---------|-----------------|
| 07 - Conclusion | 1       | ~450            |
| Total           | 18      | ~9,550          |

## Section 00: Welcome

00.0 Welcome to Python Data Types and Operators 📖 (~450 words)

**Learning Objectives:**

- Understand what this course covers and why it matters
- Recognize the connection between JavaScript types and Python types
- Set expectations for the no-AI learning approach

**Content Outline:**

- Course overview: what you'll learn about Python data types
- Why understanding types matters for backend development
- The JavaScript-to-Python bridge: familiar concepts, new syntax
- How to approach this course: write code, don't copy it

**Transitions:** This lesson sets the stage for Section 01, where we dive into Python's numeric and boolean types.

**Key Principles:**

- Data types are the building blocks of any program
- Python is dynamically typed but still type-aware
- Learning by association accelerates understanding

**Examples:**

- JavaScript `let x = 5` vs Python `x = 5` — same concept, simpler syntax
- Preview of how `typeof` in JS relates to `type()` in Python

## Section 01: Understanding Primitive Data Types

01.0 Numbers in Python: Integers and Floats 📖 (~550 words)

**Learning Objectives:**

- Distinguish between integers (int) and floating-point numbers (float)
- Identify numeric types by their visual composition
- Understand when Python automatically assigns each type

**Content Outline:**

- What is an integer in Python: whole numbers without decimals
- What is a float: numbers with decimal points
- How to recognize each type by looking at the value

- Automatic type assignment: `5` becomes int, `5.0` becomes float
- The `type()` function for verification

**Transitions:** Now that you understand numeric types, the next lesson introduces booleans—the simplest yet most powerful type for decision-making.

### Key Principles:

- Integers are exact; floats can have precision limitations
- Python decides the type based on how you write the value
- Always verify with `type()` when uncertain

### Examples:

- `age = 25` → int (no decimal point)
  - `price = 19.99` → float (has decimal point)
  - `temperature = 0.0` → float (even though value is zero, the `.0` makes it float)
- 

## 01.1 Booleans: The Logic Gates of Programming 📖 (~500 words)

### Learning Objectives:

- Understand the boolean type and its two possible values
- Recognize how booleans emerge from comparisons
- Identify truthy and falsy concepts in Python

### Content Outline:

- The boolean type: only `True` or `False` (capitalized in Python)
- Where booleans come from: comparison results
- Truthy and falsy values: what Python considers "true" or "false"
- Common falsy values: `0`, `0.0`, `""`, `None`, `False`

**Transitions:** Booleans answer yes/no questions. Next, we explore strings—the type that handles all text in your programs.

### Key Principles:

- Booleans are the foundation of all decision-making in code
- Python uses capitalized `True` and `False` (unlike JavaScript's lowercase)
- Every value in Python has a "truthiness"

### Examples:

- `is_active = True` → boolean
  - `5 > 3` → evaluates to `True`
  - `bool(0)` → `False` (zero is falsy)
  - `bool("hello")` → `True` (non-empty string is truthy)
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## 01.2 Strings: Working with Text 📖 (~550 words)

**Learning Objectives:**

- Identify strings by their quotation marks
- Understand the three ways to define strings in Python
- Recognize when to use single, double, or triple quotes

**Content Outline:**

- What is a string: any sequence of characters inside quotes
- Single quotes vs double quotes: both work identically
- Triple quotes for multi-line strings
- Empty strings: `""` or `''`
- Strings are immutable in Python

**Transitions:** You now know the four main primitive types: `int`, `float`, `bool`, and `str`. Time to put this knowledge to the test with a hands-on challenge.

**Key Principles:**

- Quotation marks are the visual identifier of strings
- Choose quote style based on content (use double if text contains single quotes)
- Triple quotes preserve line breaks

**Examples:**

- `name = "Alice"` → string (double quotes)
  - `message = 'Hello'` → string (single quotes)
  - `paragraph = """Line one\nLine two"""` → multi-line string
  - `empty = ""` → empty string (still a string, just with no characters)
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## 01.3 Identify the Data Type 🖥️ (~500 words)

**Challenge Prompt:** You will receive several variables with different values. Your task is to create a function that returns the type name as a string. This challenge tests your ability to recognize Python's primitive data types by their values.

Complete the function `identify_type(value)` that:

1. Receives any value as a parameter
2. Returns a string indicating the type: `"int"`, `"float"`, `"bool"`, `"str"`, or `"unknown"`

**Solution Code:**

```
def identify_type(value):  
    if type(value) == int:  
        return "int"  
    elif type(value) == float:  
        return "float"  
    elif type(value) == bool:  
        return "bool"
```

```

elif type(value) == str:
    return "str"
else:
    return "unknown"

# Test cases
print(identify_type(42))          # Expected: "int"
print(identify_type(3.14))       # Expected: "float"
print(identify_type(True))       # Expected: "bool"
print(identify_type("hello"))    # Expected: "str"
print(identify_type(0))          # Expected: "int"
print(identify_type(0.0))        # Expected: "float"

```

### Challenge Code:

```

def identify_type(value):
    # Your code here
    # Check the type of 'value' and return the appropriate string
    # Return "int", "float", "bool", "str", or "unknown"
    pass

# Test cases - do not modify
print(identify_type(42))          # Expected: "int"
print(identify_type(3.14))       # Expected: "float"
print(identify_type(True))       # Expected: "bool"
print(identify_type("hello"))    # Expected: "str"
print(identify_type(0))          # Expected: "int"
print(identify_type(0.0))        # Expected: "float"

```

## Section 02: The None Type and Type Detection

### 02.0 None: The Absence of Value 📖 (~500 words)

#### Learning Objectives:

- Understand what None represents in Python
- Recognize when None is used vs other falsy values
- Compare None to JavaScript's null and undefined

#### Content Outline:

- What is None: Python's way of saying "nothing here"
- None is not zero, not an empty string, not False
- Common uses: default function returns, placeholder values
- Checking for None: use `is None`, not `== None`
- JavaScript parallel: similar to `null`, Python has no `undefined`

**Transitions:** Now that you know all five primitive types, let's learn how to programmatically detect and verify types in your code.

### Key Principles:

- None is a specific type (NoneType), not just a value
- Use `is` for None comparisons, not `==`
- Functions without explicit return statements return None

### Examples:

- `result = None` → explicitly setting no value
  - `def greet(): print("Hi")` → returns None implicitly
  - `value is None` → correct way to check
  - `value == None` → works but not recommended (Pythonic style prefers `is`)
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## 02.1 Checking Types with `type()` and `isinstance()` 📖 (~550 words)

### Learning Objectives:

- Use `type()` to get the exact type of any value
- Use `isinstance()` to check if a value belongs to a type
- Understand when to use each approach

### Content Outline:

- The `type()` function: returns the exact type object
- Comparing types: `type(x) == int`
- The `isinstance()` function: checks if value is of a type (or subtypes)
- When to use each: `type()` for exact match, `isinstance()` for flexibility
- Printing types for debugging: labeled prints with `type()`

**Transitions:** With your knowledge of all primitive types and how to detect them, let's practice with a challenge that combines everything.

### Key Principles:

- `type()` returns the type object, not a string
- `isinstance()` is more flexible and handles inheritance
- Labeled prints help debug type issues: `print("x type:", type(x))`

### Examples:

- `type(5)` → `<class 'int'>`
  - `type(5) == int` → `True`
  - `isinstance(5, int)` → `True`
  - `isinstance(True, int)` → `True` (booleans are a subtype of int in Python!)
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## 02.2 Type Detective Challenge 📖 (~550 words)

**Challenge Prompt:** Create a function that analyzes a list of values and returns a summary dictionary counting how many values of each type are present. This challenge reinforces your ability to detect types programmatically.

Complete the function `count_types(values)` that:

1. Receives a list of mixed values
2. Returns a dictionary with type counts: `{"int": X, "float": X, "bool": X, "str": X, "none": X}`
3. Only count the five primitive types we've learned

### Solution Code:

```
def count_types(values):
    counts = {
        "int": 0,
        "float": 0,
        "bool": 0,
        "str": 0,
        "none": 0
    }

    for value in values:
        # Check bool BEFORE int (bool is subtype of int)
        if type(value) == bool:
            counts["bool"] += 1
        elif type(value) == int:
            counts["int"] += 1
        elif type(value) == float:
            counts["float"] += 1
        elif type(value) == str:
            counts["str"] += 1
        elif value is None:
            counts["none"] += 1

    return counts

# Test cases
test_list = [1, 2.5, "hello", True, None, 0, "", False, 3.14]
result = count_types(test_list)
print(result)
# Expected: {"int": 2, "float": 2, "bool": 2, "str": 2, "none": 1}
```

### Challenge Code:

```
def count_types(values):
    counts = {
        "int": 0,
        "float": 0,
```

```
        "bool": 0,
        "str": 0,
        "none": 0
    }

    # Your code here
    # Loop through values and count each type
    # Remember: check bool BEFORE int!

    return counts

# Test cases - do not modify
test_list = [1, 2.5, "hello", True, None, 0, "", False, 3.14]
result = count_types(test_list)
print(result)
# Expected: {"int": 2, "float": 2, "bool": 2, "str": 2, "none": 1}
```

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## Section 03: Arithmetic and Assignment Operators

### 03.0 Arithmetic Operators: Math in Python 📖 (~550 words)

#### Learning Objectives:

- Use all arithmetic operators in Python
- Understand integer division vs true division
- Apply the modulo operator for remainders

#### Content Outline:

- Basic operators: `+`, `-`, `*`, `/`
- True division `/` always returns float
- Integer division `//` returns int (floors the result)
- Modulo `%` returns the remainder
- Exponentiation `**` for powers
- Operator precedence: PEMDAS applies

**Transitions:** Now that you can perform calculations, let's learn how to store and update values efficiently with assignment operators.

#### Key Principles:

- Division in Python 3 always returns float (unlike Python 2)
- Use `//` when you need whole number results
- Modulo is essential for cycles, even/odd checks, and wrapping

#### Examples:

- `10 / 3` → `3.333...` (float)
- `10 // 3` → `3` (int, floors toward negative infinity)

- `10 % 3` → 1 (remainder)
  - `2 ** 10` → 1024 (2 to the power of 10)
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### 03.1 Assignment Operators: Storing and Updating Values 📖 (~500 words)

#### Learning Objectives:

- Use basic assignment with `=`
- Apply compound assignment operators
- Understand the difference between assignment and comparison

#### Content Outline:

- Basic assignment: `x = 5`
- Compound operators: `+=`, `-=`, `*=`, `/=`, `//=`, `%=`, `**=`
- How compound operators work: `x += 1` is shorthand for `x = x + 1`
- Multiple assignment: `a, b = 1, 2`
- Common mistake: confusing `=` (assignment) with `==` (comparison)

**Transitions:** With arithmetic and assignment mastered, let's put them together in a practical calculator challenge.

#### Key Principles:

- `=` assigns, `==` compares
- Compound operators modify in place (conceptually)
- Multiple assignment enables clean swaps: `a, b = b, a`

#### Examples:

- `score = 0` then `score += 10` → score is now 10
  - `price = 100` then `price *= 0.9` → price is now 90.0 (10% discount)
  - `x, y = 5, 10` → assigns 5 to x and 10 to y simultaneously
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### 03.2 Build a Simple Calculator 🖥️ (~550 words)

**Challenge Prompt:** Create a function that performs basic arithmetic operations. The function receives two numbers and an operator string, then returns the result. Handle division by zero gracefully.

Complete the function `calculate(a, b, operator)` that:

1. Receives two numbers and an operator string ("`+`", "`-`", "`*`", "`/`", "`//`", "`%`", "`**`")
2. Returns the result of the operation
3. Returns `None` if division by zero is attempted
4. Returns `None` if operator is not recognized

#### Solution Code:

```
def calculate(a, b, operator):
    if operator == "+":
        return a + b
    elif operator == "-":
        return a - b
    elif operator == "*":
        return a * b
    elif operator == "/":
        if b == 0:
            return None
        return a / b
    elif operator == "//":
        if b == 0:
            return None
        return a // b
    elif operator == "%":
        if b == 0:
            return None
        return a % b
    elif operator == "**":
        return a ** b
    else:
        return None

# Test cases
print(calculate(10, 3, "+"))    # Expected: 13
print(calculate(10, 3, "-"))    # Expected: 7
print(calculate(10, 3, "*"))    # Expected: 30
print(calculate(10, 3, "/"))    # Expected: 3.333...
print(calculate(10, 3, "//"))   # Expected: 3
print(calculate(10, 3, "%"))    # Expected: 1
print(calculate(2, 10, "**"))    # Expected: 1024
print(calculate(10, 0, "/"))    # Expected: None
print(calculate(10, 3, "^"))    # Expected: None (invalid operator)
```

### Challenge Code:

```
def calculate(a, b, operator):
    # Your code here
    # Perform the operation based on the operator string
    # Handle division by zero (return None)
    # Handle unknown operators (return None)
    pass

# Test cases - do not modify
print(calculate(10, 3, "+"))    # Expected: 13
print(calculate(10, 3, "-"))    # Expected: 7
print(calculate(10, 3, "*"))    # Expected: 30
print(calculate(10, 3, "/"))    # Expected: 3.333...
```

```
print(calculate(10, 3, "//")) # Expected: 3
print(calculate(10, 3, "%")) # Expected: 1
print(calculate(2, 10, "**")) # Expected: 1024
print(calculate(10, 0, "/")) # Expected: None
print(calculate(10, 3, "^")) # Expected: None (invalid operator)
```

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## Section 04: Comparison and Logical Operators

### 04.0 Comparison Operators: Evaluating Conditions 📖 (~550 words)

#### Learning Objectives:

- Use all comparison operators in Python
- Understand the difference between `==` and `is`
- Apply comparisons to different data types

#### Content Outline:

- Equality: `==` (equal) and `!=` (not equal)
- Ordering: `<`, `>`, `<=`, `>=`
- Identity: `is` and `is not` (same object in memory)
- Comparison results are always boolean
- String comparisons: lexicographic (alphabetical) order
- Type matters: `"5" == 5` is `False`

**Transitions:** Individual comparisons are useful, but real conditions often combine multiple checks. Next, we explore logical operators.

#### Key Principles:

- `==` checks value equality, `is` checks identity (same object)
- Comparisons always return `True` or `False`
- Python allows chaining: `1 < x < 10` (unlike JavaScript)

#### Examples:

- `5 == 5` → `True`
- `5 == 5.0` → `True` (value is same, type coercion happens)
- `"5" == 5` → `False` (string vs int, no coercion)
- `x = [1, 2]; y = [1, 2]; x == y` → `True` (same values)
- `x is y` → `False` (different objects in memory)

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### 04.1 Logical Operators: Combining Conditions 📖 (~500 words)

#### Learning Objectives:

- Use `and`, `or`, and `not` operators
- Understand short-circuit evaluation
- Combine multiple conditions effectively

## Content Outline:

- The **and** operator: both must be True
- The **or** operator: at least one must be True
- The **not** operator: inverts the boolean
- Short-circuit evaluation: Python stops early when result is determined
- Operator precedence: **not** > **and** > **or**
- JavaScript parallel: **&&**, **||**, **!** → **and**, **or**, **not**

**Transitions:** Let's solidify your understanding of comparisons and logical operators with a truth table challenge.

## Key Principles:

- **and** returns the first falsy value or the last value
- **or** returns the first truthy value or the last value
- Use parentheses to clarify complex conditions

## Examples:

- **True and False** → **False**
  - **True or False** → **True**
  - **not True** → **False**
  - **5 > 3 and 10 < 20** → **True**
  - **x = 0; y = x or "default"** → y is "default" (short-circuit)
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## 04.2 Truth Table Challenge 📖 (~550 words)

**Challenge Prompt:** Create functions that evaluate complex conditions using comparison and logical operators. This challenge tests your ability to combine operators correctly.

Complete the following functions:

1. **is\_valid\_age(age)**: Returns **True** if age is between 18 and 65 (inclusive)
2. **is\_weekend(day)**: Returns **True** if day is "Saturday" or "Sunday"
3. **can\_access(age, has\_permission)**: Returns **True** if age >= 18 AND has\_permission is True
4. **needs\_review(score)**: Returns **True** if score is less than 60 OR greater than 95

## Solution Code:

```
def is_valid_age(age):
    return age >= 18 and age <= 65

def is_weekend(day):
    return day == "Saturday" or day == "Sunday"

def can_access(age, has_permission):
    return age >= 18 and has_permission
```

```
def needs_review(score):
    return score < 60 or score > 95

# Test cases
print(is_valid_age(25))      # Expected: True
print(is_valid_age(17))      # Expected: False
print(is_valid_age(65))      # Expected: True

print(is_weekend("Saturday")) # Expected: True
print(is_weekend("Monday"))   # Expected: False

print(can_access(20, True))   # Expected: True
print(can_access(20, False))  # Expected: False
print(can_access(15, True))   # Expected: False

print(needs_review(55))       # Expected: True
print(needs_review(75))       # Expected: False
print(needs_review(98))       # Expected: True
```

### Challenge Code:

```
def is_valid_age(age):
    # Return True if age is between 18 and 65 (inclusive)
    pass

def is_weekend(day):
    # Return True if day is "Saturday" or "Sunday"
    pass

def can_access(age, has_permission):
    # Return True if age >= 18 AND has_permission is True
    pass

def needs_review(score):
    # Return True if score < 60 OR score > 95
    pass

# Test cases - do not modify
print(is_valid_age(25))      # Expected: True
print(is_valid_age(17))      # Expected: False
print(is_valid_age(65))      # Expected: True

print(is_weekend("Saturday")) # Expected: True
print(is_weekend("Monday"))   # Expected: False
```

```
print(can_access(20, True))    # Expected: True
print(can_access(20, False))   # Expected: False
print(can_access(15, True))    # Expected: False

print(needs_review(55))        # Expected: True
print(needs_review(75))        # Expected: False
print(needs_review(98))        # Expected: True
```

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## Section 05: Common Mistakes with Types and Operators

### 05.0 Anti-Patterns: Type Confusion and Operator Misuse 📖 (~550 words)

#### Learning Objectives:

- Recognize common type-related mistakes
- Avoid operator confusion pitfalls
- Apply correct patterns instead of anti-patterns

#### Content Outline:

- Anti-pattern 1: Comparing different types expecting coercion (`"5" == 5`)
- Anti-pattern 2: Using `=` when you mean `==`
- Anti-pattern 3: Checking `== True` or `== False` instead of using the value directly
- Anti-pattern 4: Not handling `None` before operations
- Anti-pattern 5: Integer division surprise (expecting float, getting int)
- Anti-pattern 6: Modifying strings expecting mutation

**Transitions:** Now that you know what NOT to do, let's practice identifying and fixing these mistakes in real code.

#### Key Principles:

- Python doesn't coerce types in comparisons like JavaScript does
- Trust boolean values directly: `if is_valid:` not `if is_valid == True:`
- Always consider `None` as a possible value

#### Examples:

✗ Anti-pattern: Type coercion expectation

```
user_input = "5"
if user_input == 5: # Always False!
    print("Match")
```

☒ Correct:

```
user_input = "5"
if int(user_input) == 5:
    print("Match")
```

✗ Anti-pattern: Verbose boolean check

```
if is_active == True:
    do_something()
```

☑ Correct:

```
if is_active:
    do_something()
```

✗ Anti-pattern: Assignment instead of comparison

```
if x = 5: # SyntaxError in Python (thankfully!)
    print("x is 5")
```

☑ Correct:

```
if x == 5:
    print("x is 5")
```

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## 05.1 Fix the Broken Code 🖨️ (~500 words)

**Challenge Prompt:** The following code contains multiple type and operator errors. Your task is to fix all the bugs so the function works correctly. Each bug relates to an anti-pattern covered in the previous lesson.

The function `validate_user(name, age, is_active)` should:

1. Return `"Invalid name"` if name is None or empty string
2. Return `"Invalid age"` if age is not an integer or is negative
3. Return `"Inactive user"` if `is_active` is False
4. Return `"Valid user: [name]"` if all checks pass

**Solution Code:**

```
def validate_user(name, age, is_active):
    # Check for None or empty name
    if name is None or name == "":
```

```

        return "Invalid name"

    # Check if age is integer and non-negative
    if type(age) != int or age < 0:
        return "Invalid age"

    # Check if user is active
    if not is_active:
        return "Inactive user"

    return "Valid user: " + name

# Test cases
print(validate_user(None, 25, True))      # Expected: "Invalid name"
print(validate_user("", 25, True))        # Expected: "Invalid name"
print(validate_user("Alice", "25", True)) # Expected: "Invalid age"
print(validate_user("Alice", -5, True))   # Expected: "Invalid age"
print(validate_user("Alice", 25, False))  # Expected: "Inactive user"
print(validate_user("Alice", 25, True))   # Expected: "Valid user: Alice"

```

### Challenge Code:

```

def validate_user(name, age, is_active):
    # BUG 1: Wrong None check
    if name = None or name = "":
        return "Invalid name"

    # BUG 2: Missing type check for age
    if age < 0:
        return "Invalid age"

    # BUG 3: Verbose boolean check
    if is_active == False:
        return "Inactive user"

    # BUG 4: Type coercion assumption
    return "Valid user: " + name

# Test cases - do not modify
print(validate_user(None, 25, True))      # Expected: "Invalid name"
print(validate_user("", 25, True))        # Expected: "Invalid name"
print(validate_user("Alice", "25", True)) # Expected: "Invalid age"
print(validate_user("Alice", -5, True))   # Expected: "Invalid age"
print(validate_user("Alice", 25, False))  # Expected: "Inactive user"
print(validate_user("Alice", 25, True))   # Expected: "Valid user: Alice"

```

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## Section 06: Assessment

## 06.0 Knowledge Check 🧠 (~700 words)

### Learning Objectives:

- Demonstrate understanding of Python primitive data types
- Apply operators correctly in various scenarios
- Identify type-related issues in code

### Question Format: Multiple Choice

### Questions:

**Question 1:** What is the type of the value `3.0` in Python?

- A) int
- B) float
- C) str
- D) number

**Correct Answer:** B) float **Explanation:** Any number with a decimal point is a float in Python, even if the decimal part is zero.

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**Question 2:** What does `10 // 3` return in Python?

- A) 3.333...
- B) 3
- C) 4
- D) 3.0

**Correct Answer:** B) 3 **Explanation:** The `//` operator performs integer division (floor division), returning the largest integer less than or equal to the result.

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**Question 3:** What is the result of `"5" == 5` in Python?

- A) True
- B) False
- C) "5"
- D) Error

**Correct Answer:** B) False **Explanation:** Python does not perform automatic type coercion in comparisons. A string and an integer are never equal.

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**Question 4:** Which operator should you use to check if a variable is None?

- A) `== None`
- B) `= None`
- C) `is None`
- D) `equals(None)`

**Correct Answer:** C) `is None` **Explanation:** The `is` operator checks identity and is the Pythonic way to check for None.

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**Question 5:** What does `True and False or True` evaluate to?

- A) True
- B) False
- C) None
- D) Error

**Correct Answer:** A) True **Explanation:** `and` has higher precedence than `or`. So: `(True and False) or True` → `False or True` → `True`.

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**Question 6:** What is the result of `type(True) == int`?

- A) True
- B) False
- C) Error
- D) None

**Correct Answer:** B) False **Explanation:** `type(True)` returns `<class 'bool'>`, not `<class 'int'>`. However, `isinstance(True, int)` would return True because bool is a subclass of int.

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**Question 7:** What value does a function return if it has no return statement?

- A) 0
- B) False
- C) ""
- D) None

**Correct Answer:** D) None **Explanation:** Functions without an explicit return statement implicitly return None.

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#### Evaluation Criteria:

- 7/7 correct: Excellent understanding, ready for control flow
  - 5-6/7 correct: Good grasp, review missed concepts
  - 3-4/7 correct: Review sections related to missed questions
  - 0-2/7 correct: Revisit the entire course before proceeding
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## Section 07: Conclusion

### 07.0 Next Steps: From Types to Flow Control (~450 words)

#### Content Outline:

**What You Accomplished:** You've built a solid foundation in Python's type system. You can now identify integers, floats, booleans, strings, and None by sight. You understand how arithmetic, assignment,

comparison, and logical operators work—and importantly, how they differ from JavaScript in subtle but significant ways.

**The Bigger Picture:** Data types and operators are the atoms of programming. Every program you write—from simple scripts to complex backend systems—manipulates data using these fundamental building blocks. The comparison and logical operators you learned today are the gateway to the next skill: control flow.

**What's Coming Next:** In the next learnpack, you'll learn to control the flow of your programs using:

- Conditionals: `if`, `elif`, `else`
- Loops: `for`, `while`
- Flow interrupts: `break`, `continue`

The boolean results from your comparison operators will drive these decisions. The `and`, `or`, and `not` operators will combine conditions in powerful ways.

### Keep Practicing:

- Write code by hand—don't copy and paste
- Use `print()` with labels to debug: `print("age type:", type(age))`
- When in doubt, check the type: `type(x)`
- Research syntax when needed—memorization comes with practice

### Resources for Further Exploration:

- Python official documentation: [docs.python.org](https://docs.python.org)
- Practice exercises: Continue with the Mastering Python exercises
- When stuck: Ask Rigobot for hints, but write your own solutions

**A Note on the No-AI Rule:** This course restricts AI assistance intentionally. Writing code yourself—even struggling with it—builds neural pathways that copy-pasting never will. The goal isn't to finish fast; it's to understand deeply. Trust the process.

You're ready for control flow. Let's keep building.

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